



Interactive Keyboard: Final Critique

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Project Description

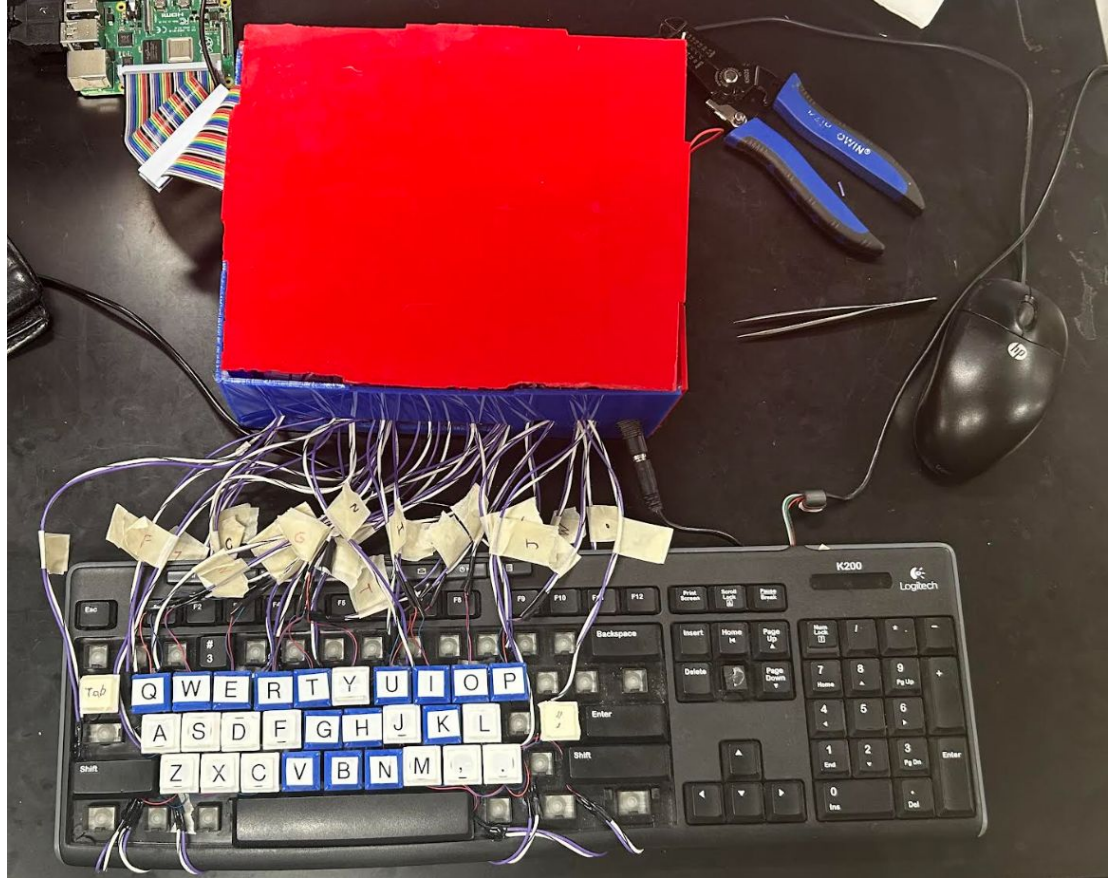
We designed a keyboard intended to enhance typing education through dynamic tactile feedback. Unlike conventional static keyboards, this interface can actively vibrate keys to reinforce correct finger placement and provide real-time haptic feedback. The goal of this project is to test enhanced input and output mechanisms to increase human computer interaction capabilities.

During our design process, we developed and tested stepper motor, haptic motor, and pneumatic actuation feedback mechanisms to determine their effectiveness, scalability, and integration complexity. Manufacturing methods included 3D printing custom keycaps, circuit design utilizing a variety of motor types, PCB design, silicon molding, and dynamic LED integration.

Ultimately, we decided to make a fully sized haptic motor based keyboard that utilizes a raspberry pi to record user inputs and provide user input. We will metrics of typing speed, typing accuracy, and qualitative user feedback to determine the effectiveness of our design.

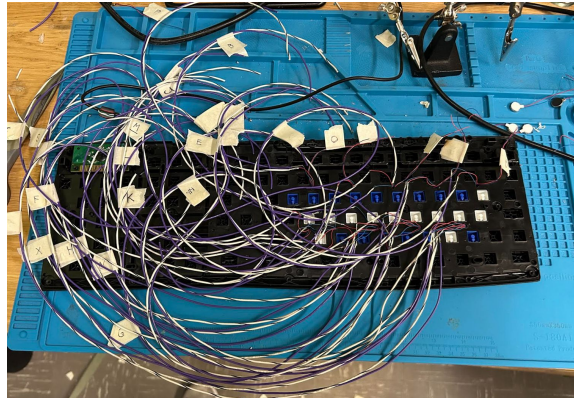
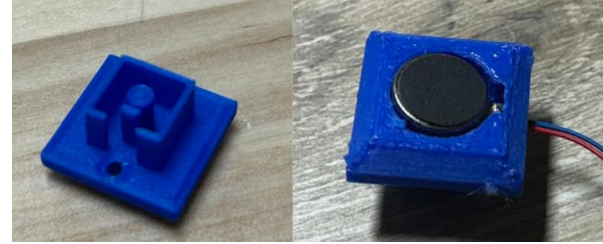
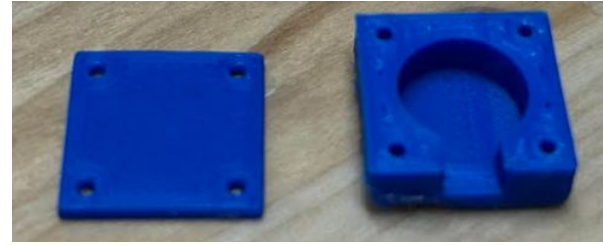
Final Product

For our final product this semester, we created a prototype, haptic vibration keyboard, with 26 vibrating letter keys, a vibrating comma key, and a vibrating period key. We also developed a software which incorporates a fun, interactive typing game that vibrates specific keys on command to assist the user in determining which key to press.



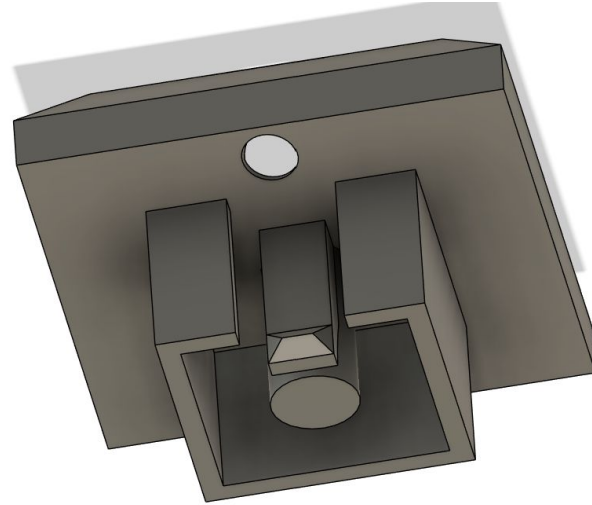
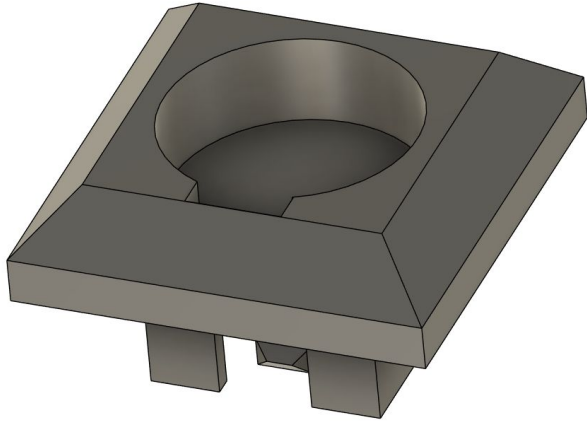
Iterative Prototyping

- For the individual keys, our prototypes went from simple 3D printed cases for the haptic motor to specially designed key inserts, which fit into the the logitech keyboard and could house the motors
- Initial motor wiring ran underneath the top plastic layer of the keyboard, but we ended up running the wires through the top of the keyboard in between the key gaps in order to not interfere with the pressure sensitive mesh inside of the keyboard

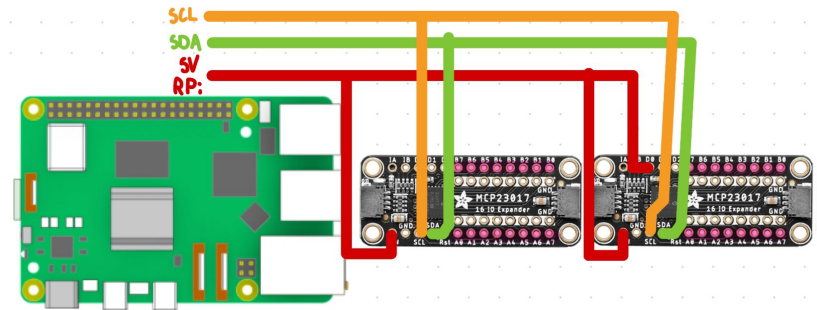




CAD Models



Wiring/Electronics

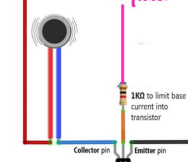


5V Power Supply

DIO of IO expander

GND

x 29



Programming

- Motor utilities program allows for activation, testing, and key mapping of 26+ vibrating keys
- Using Pygame, a simple typing game was created to prompt and record keyboard input and output vibrations to the haptic motors
- Simple mode that includes single key prompts to test basic typing skills
- Book mode that includes random excerpts from Public Domain children books downloaded from Project Gutenberg
- Metrics of words per minute, accuracy, most missed key, and excerpt length/difficulty are recorded into a CSV file for later processing





Engineering Standards

Ergonomics and Human Factors

(FAA HFDS / ISO 9241)

- Wrist extension and ulnar deviation $< 15^\circ$
- Keyboard slope $\leq 15^\circ$
- Matte, non-reflective keyboard surface
- Smooth enclosure edges for user safety

Keyboard Geometry

(FAA HFDS Keyboard Guidelines)

- Key strike surface width: 12–15 mm
- Horizontal spacing: 16–19 mm center-to-center
- Vertical spacing: 18–21 mm
- Square/rectangular key shape preferred
- Standard QWERTY/numpad-inspired layout

Key Performance & Haptics

(FAA HFDS / Haptic Interaction Principles)

- Key actuation force: 0.25–1.5 N
- Key travel target: 2–4 mm preferred
- Consistent tactile feedback across keys
- Positive feedback required after keypress
- Key stability: no wobbling or sticking

Human-Centered Design

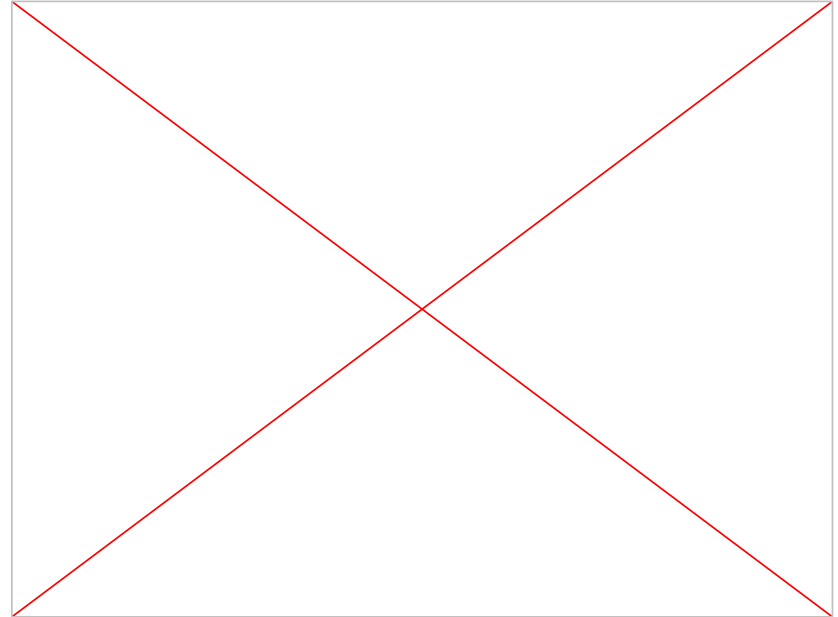
(ISO 13407)

- Iterative prototyping and testing
- User-centered interaction design
- Accessibility-focused feedback systems
- Primary school and survey-based user evaluation



Testing: Internal Evaluation Procedure

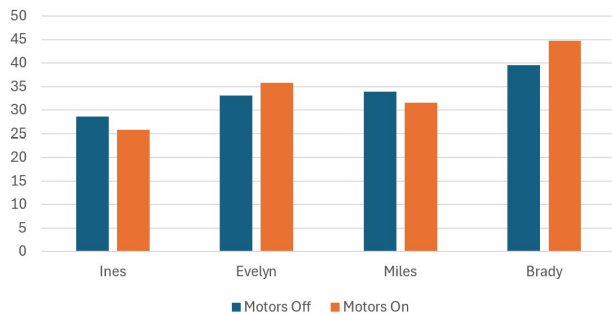
- For our internal evaluation of the final prototype, each member of our group played 2 runs of the “Book Mode” version of the typing game.
- In the first run, we conducted the game with the motors off, and in the second run, we conducted the game with the motors on.
- Using our software, we then collected data on our typing speed, accuracy, and reaction times.



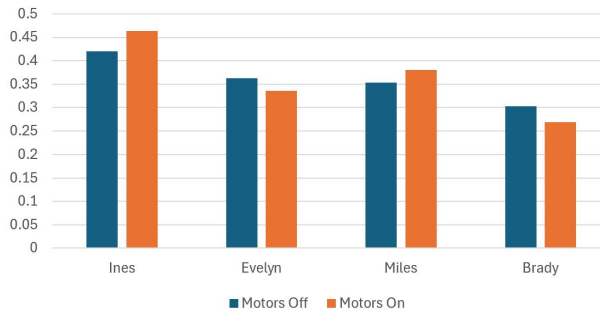


Testing: Internal Evaluation Results

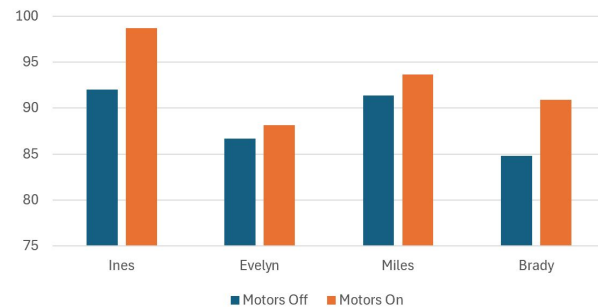
Words per Minute



Average Reaction Time

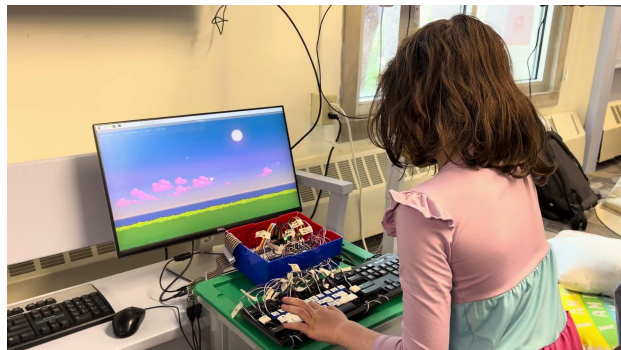
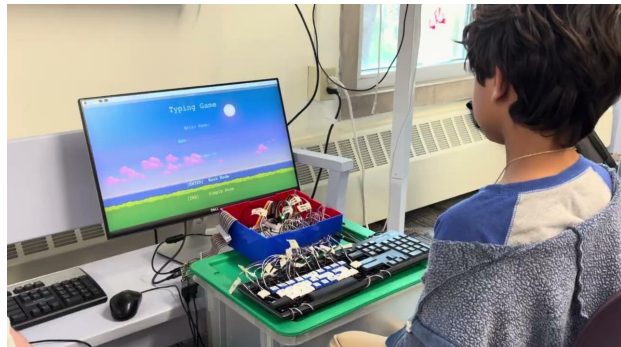
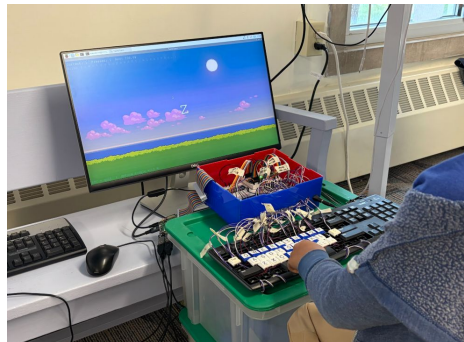


Percent Accuracy



Testing: Wheeler School Evaluation Procedure

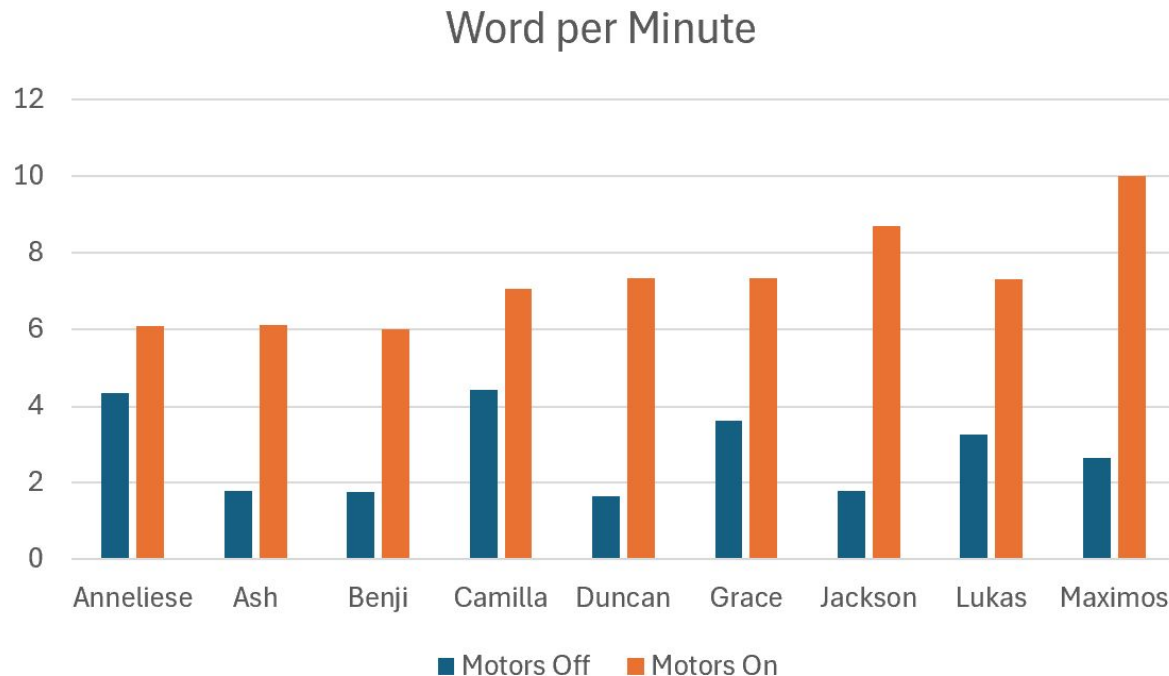
- For our testing at the Wheeler School, we had a class of 9 1st graders play the “Simple Mode” of our typing game.
- In the first run, we had them conduct the game with the motors off, and in the second run, we had them conduct the game with the motors on.
- Using our software, we then collected data on our typing speed, accuracy, and reaction times.
- After each student was done playing the games, we asked them a few questions about their thoughts on the keyboard.



Testing: Results

- Average WPM Motors Off: 2.087
- Average WPM Motors Off: 7.314
- Average WPM Increase: 4.507

Using the keyboard with the vibrational motors on resulted in a %160.56 increase in typing speed.

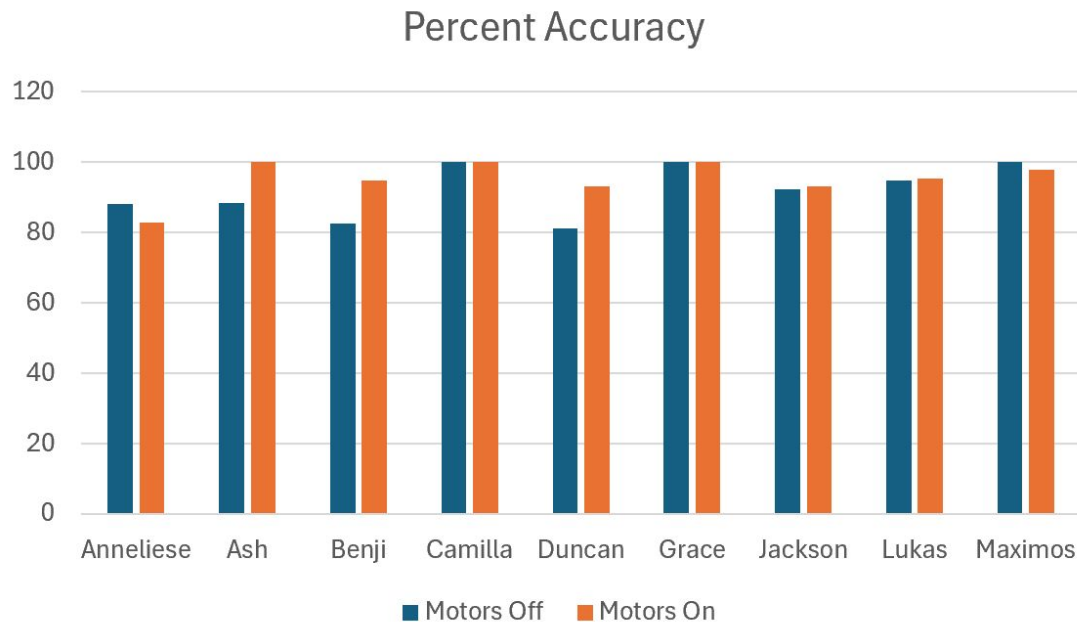




Testing: Results

- Average % Accuracy Motors Off: 91.906
- Average % Accuracy Motors On: 95.188
- Average % Accuracy Motors Off: 3.282

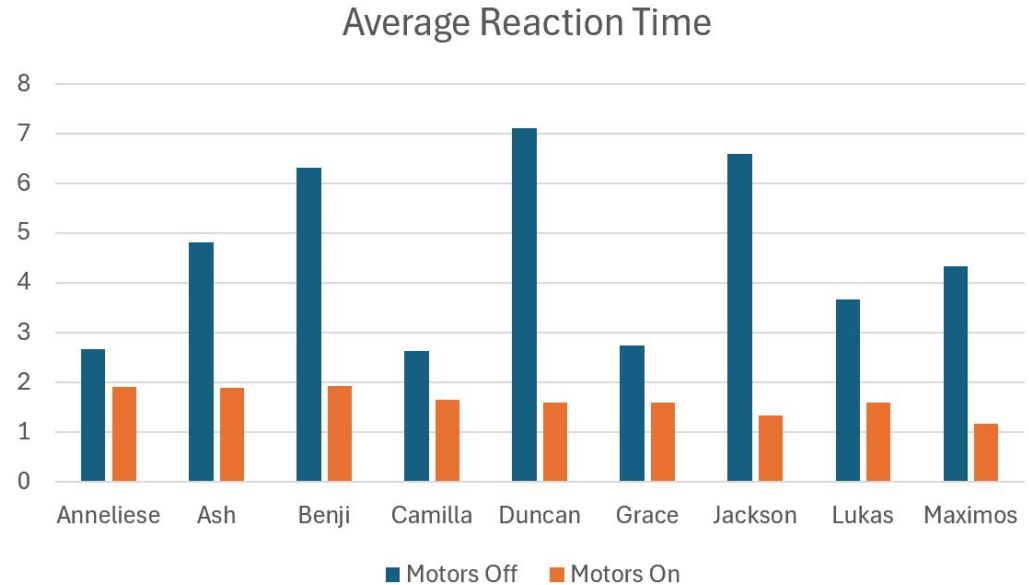
Using the keyboard with the vibrational motors on resulted in a %3.57 increase in typing accuracy.



Testing: Results

- Average Reaction Time Motors Off: 4.541
- Average % Accuracy Motors Off: 1.633
- Average % Accuracy Motors On: 2.908

Using the keyboard with the vibrational motors on resulted in a %64.03 decrease in typing reaction time.





Testing: Survey Results

Did the vibration help?	Did you like the moving/vibrating keys?	Was it fun?
Kind of	It was a little much	Yes
Yes	Liked it when they were moving around, it was eas	Yes
Yes	A little bit much	Yes
Yes	Like it	Yes
Yes	Liked it	Yes
Yes	Liked it, kind of showed what to push	Yes
Yes, can know which letter immedi	Liked it	Yes (enthusiastic)
Yes	Liked it, but a little shaky	Yes
Yes	Yes, liked it	Yes, very fun



Complications

- Lose Keys: Some of the keys were not fastened to the keyboard securely enough and the vibration of the motors would make them pop out of the keyboard
- Motors Overheating: When left vibrating for a long period of time, some of the haptic vibration motors would overheat and become warm or even hot to the touch
- Students Looking Down: When the motors were activated, some of the students would ignore the screen completely, and use the visible vibration of the keys to know which key to press
- Key Malfunctions: Some of the wiring on the breadboard was imperfect, and on occasion, the motor wires, transistors, or resistors would disconnect from the breadboard, making some of the vibration keys inactive.



Design Impact

Economic	Global keyboard market is valued from hundred of millions to multiple billion USD. We explored a low-cost and scalable approach to interactive typing education, which can support affordable educational and assistive technology for schools, learning centers, and accessibility-focused applications.
Social	We aimed to make typing education more interactive and accessible through tactile guidance and real-time feedback. The system may support users with different learning styles, beginners developing typing skills, and individuals who benefit from assistive or non-visual feedback systems.



Design Impact

Environmental	Use of compact embedded electronics and low-power haptic motors to minimize energy consumption during operation. Future iterations will further evaluate material selection, manufacturability, repairability, and long-term sustainability considerations.
Global	A scalable interactive keyboard platform could support typing education and accessibility across different learning environments, languages, and user needs worldwide.

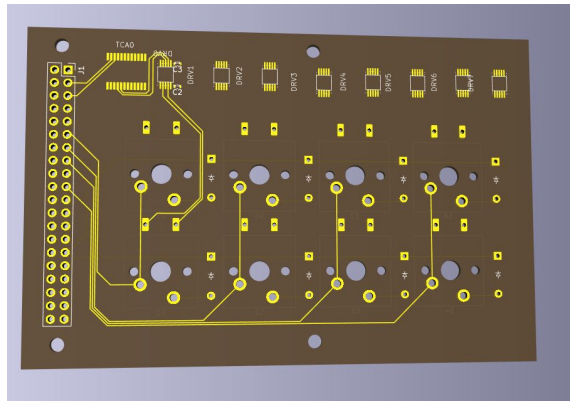


Future Iterations and Goals

- Add Keys: The next iteration of the keyboard should be include vibration keys for the space bar, number keys, shift key, and enter key
- Improve Keys: Improve the design of the keys as well as the material used to produce them in order to prevent the keys from flying out of the keyboard; this will also prevent the keys from moving around in the keyboard as much, which should prevent the students from using the motion of the vibrating keys to know which keys to press
- Finger Placement: The next keyboard prototype should include pressure sensitive pads on the keys, such that the keyboard will not activate until the student has their fingers in the correct orientation on the keyboard

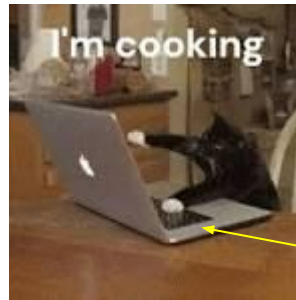
Future Iterations and Goals

- Small PCB Assembly: Assembly of current, 8 key PCB design can lead to a more precisely controlled, scalable, more robust system with LED and haptic integration
- PCB expansion: Modular expansion to 36 keys + PCB reprint will allow for a mass manufacturable, final keyboard design that could function as a final product.





Thank You!!



Reactive Keyboard